

# Full Big-Company SaaS Development Process

Project Team

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## 1 Phase 0 — Vision and Product Framing

### 1.1 Problem Statement

- What problem exists today?
- Why do current solutions fail?
- Who experiences the problem?

**Example:** Utilities struggle to adapt electricity distribution in real time due to mixed energy sources and fragile grid connectivity.

### 1.2 Value Proposition

A decision-support platform that models grid connectivity and recommends optimal energy dispatch under failures and renewable variability.

### 1.3 Target Users

- Grid operators
- Utility planners
- Energy regulators (read-only)

### 1.4 Success Metrics (KPIs)

- Blackout duration reduction
- Renewable utilization percentage
- Decision latency
- Simulation accuracy

## **2 Phase 1 — Requirements Engineering**

### **2.1 Stakeholder Analysis**

- Utility company
- Government
- Operators
- Engineers

### **2.2 Functional Requirements**

- Model grid as a graph
- Detect disconnected components
- Simulate failures
- Recommend energy source switching

### **2.3 Non-Functional Requirements**

- High availability
- Deterministic decisions
- Explainability
- Scalability

### **2.4 Constraints**

- No real IoT input (simulated data only)
- Limited historical data
- Regulatory assumptions

## **3 Phase 2 — Domain Modeling**

### **3.1 Core Domain Concepts**

- Node
- Edge
- Component
- PowerPlant
- Load
- Event (failure, recovery)

### **3.2 Domain Rules**

- A component is alive if it has at least one active power plant
- Renewable energy has priority over thermal
- Hospitals must always be served first

### **3.3 State Transitions**

Normal  $\rightarrow$  Failure  $\rightarrow$  Islanded  $\rightarrow$  Recovery

## **4 Phase 3 — System Architecture**

### **4.1 High-Level Architecture**

- Core Decision Engine
- Simulation Engine
- API Layer
- Web Dashboard

### **4.2 Architectural Style**

- Modular monolith (early SaaS)
- API-first design
- Stateless services

## **5 Phase 4 — Detailed Design**

### **5.1 Data Structure Mapping**

- Grid: Graph
- Connectivity: Union-Find
- Dispatch: Priority Queue
- Load Analysis: Segment Tree

### **5.2 Algorithm Design**

- Failure detection
- Component recalculation
- Dispatch logic
- Load shedding logic

## **6 Phase 5 — Implementation**

- Clean architecture
- Domain-driven boundaries
- Tests with every feature

## **7 Phase 6 — Testing and Validation**

- Unit testing
- Integration testing
- Scenario-based testing

## **8 Phase 7 — Documentation**

- Technical documentation
- Product documentation
- Academic report

## **9 Phase 8 — Deployment and SaaS Readiness**

- Development, staging, and production environments
- CI/CD pipelines
- Monitoring and metrics

## **10 Phase 9 — Security and Compliance**

- Authentication
- Authorization
- Audit logs
- Data integrity

## **11 Phase 10 — Post-Launch Evolution**

- Feature backlog
- Feedback loop
- Roadmap
- Future IoT integration